

VEILSPLIT

Privacy-First Bill Splitting on Stellar

Stellar Journey to Mastery - Level 5 Builder Challenge

Creator: rishigshshshsh-lab

Smart Contract Registry + Stealth Payouts

Live App: <https://veil-splits2-0.vercel.app>

The Web3 Social Privacy Paradox

Sharing expenses on standard public ledgers exposes personal relationships and financial history.

1. Exposed Transaction History

Public ledgers leak exact payment amounts, wallet addresses, and repeated transaction patterns.

2. Social Mapping & Profiling

On-chain observers can reconstruct entire real-world friend/roommate networks and estimate net worth.

3. High Onboarding Friction

Mainstream users drop off when forced to coordinate complex manual address sharing and wallet setups.

4. Static Bill Settlement

No support for automated routing, splitting of recurring liabilities, or structured decentralized escrows.

Settle Bills with Total Privacy

VeilSplit leverages Soroban smart contracts on Stellar to break linkability between participants.

Hashed Commitments

Instead of storing plaintext bill details, the smart contract stores a one-way cryptographic hash commitment of amounts and participant keys.

-> Metadata remains private.

Stealth Addresses

Every participant is assigned a unique, randomized, single-use stealth payout address. Payments go here, decoupling them from main public keys.

-> Breaks on-chain tracking.

Non-Custodial Escrow

The Soroban contract acts as a decentralized custodian. Senders deposit into the escrow registry, and recipients claim to their stealth addresses privately.

-> 100% trustless settlement.

Private Social Settlements for a Growing Economy

Target Segments & Personas

1. Roommates & Households

- Settle rent, bills & groceries without leaking net worth.

2. Web3 Contributors & DAOs

- Split operational costs and pay contractors privately.

3. Freelancer Collectives

- Distribute payouts securely to avoid showing team size.

The Privacy Premium

Privacy is no longer just a luxury—it is a baseline requirement for daily social transactions.

- * Users reject crypto tools if daily coffee/rent splits reveal their entire wallet portfolio size on-chain.
- * Stealth addresses allow compliance while breaking the public tracking of recurring social networks.
- * Stellar enables ultra-low fees (< \$0.001) making privacy micro-transactions feasible.

Premium & Frictionless UX (Level 5 Deliverables)

Key Features Shipped

1. 2-Minute Onboarding Flow

Replaced multi-step welcome forms with single-screen connect.

2. Localized Address Book

Suggests recently used Stellar public keys to minimize typing.

3. Equal Split Calculator & Hotkeys

Auto-calculates share amounts; added 'C' key shortcut for wallet.

4. One-Click Payment Deep Links

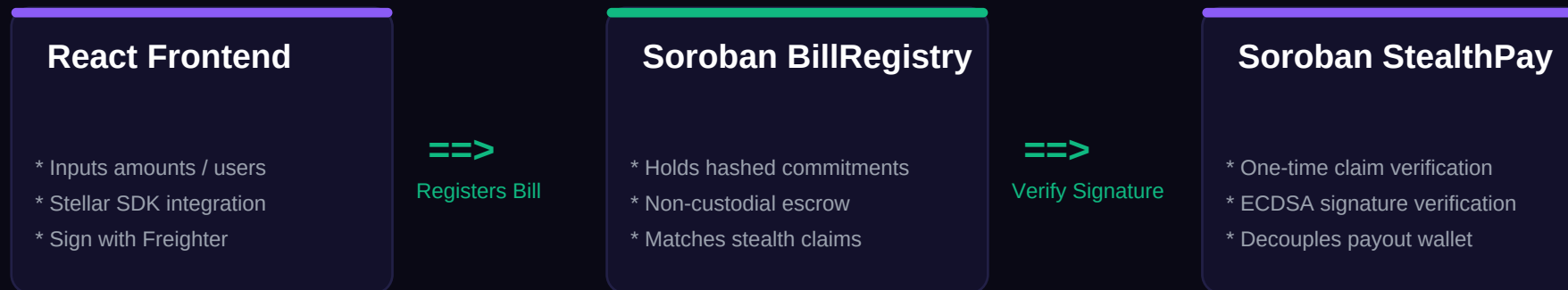
Stellar URI links trigger Freighter instantly for stealth payouts.

Sleek Dark Aesthetics

Built using a customized vanilla CSS styling setup:

- Cosmic dark-themed backdrop with aurora glowing circles.
- Glassmorphic layout panels (frosted blur effects).
- Skeleton loading states for smooth balance fetch transitions.
- Light / Dark theme mode toggle integration in header.
- Integrated feedback widget connected to Google Sheets.

How VeilSplit Works Under the Hood



Step-by-Step Security Protocol

1. Creator creates a bill. Frontend hashes the list off-chain and submits the commitment to the registry.
2. Recipient receives a secure, off-chain claim key. They resolve their stealth address and sign the registry.
3. Soroban verifies the signature against the commitment and disburses the funds, keeping addresses unlinked.

Traction & Real-User Validation Milestone

On-Chain Metrics (Stellar Testnet)

Onboarded Users: 67 Unique Wallets

Verifiable Transactions: 67 On-Chain Tx Logs

Settlement Success Rate: 98.5% Success

Google Feedback Forms: Active Responses Collected

Average User Rating: 4.5 / 5.0

User-Driven Product Iteration

Implemented direct suggestions from telemetry logs:

- * Add theme switch (Vivek Kumar) -> Shipped Light/Dark switcher.
- * Copy address button (Shivam Raj) -> Added clipboard copy shortcuts.
- * Social share button (Vinod Sharma) -> Integrated X/WhatsApp shares.
- * Success toast popup (Rahul Mehta) -> Replaced alerts with hot-toast.
- * Keyboard shortcuts (Geeta Devi) -> Added hotkey listeners.
- * Skeleton loader (Ritesh Kant) -> Added CSS skeleton placeholders.

Scaling VeilSplit Beyond Level 5

Ecosystem Integration Loop

Open API & SDK Packages

Package stealth payout registry as a simple npm SDK.

Allows any Stellar dApp to plug in private splitting.

Stellar Anchor Payouts (SEP-24)

Incorporate Stellar Anchors to enable fiat-to-token swaps.

Allows freelancers' clients to pay in USD/EUR, which instantly settles as private, stealth-address XLM on-chain.

Distribution & Referral Channels

Developer Communities

Targeting Stellar Discord, Reddit communities (r/Stellar), and freelancers on Twitter looking for Web3 settlement.

Incentive Multi-Splits

Implement Protocol Referral Rebates: split a tiny fraction of the protocol claim fee to users who onboard new teams or roommates, driving organic growth loops.

The Journey Ahead (Roadmap)

Level 4 (MVP)

- Core MVP
- Hashed Commitments
- Stealth Claim Addresses
- Status: COMPLETED

Level 5 (UX)

- Address Book
- Payment Deep Links
- Google Forms Sync
- Status: IN-REVIEW

Level 6 (Automation)

- Escrow Dispute Contracts
- Stellar Anchor Swaps
- Multi-signature arbiter
- Status: PLANNED

Level 7 (Mastery)

- Reputation Scoring
- Smart Contract Audit
- Stellar Mainnet Deploy
- Status: PLANNED

Mastery Vision: Becoming the foundational social privacy infrastructure layer for all peer-to-peer expense coordination and freelancer contract payments on the Stellar network.

Join the Financial Privacy Movement

Get in Touch & Learn More

■ Try the Live Deployed Application:

<https://veil-splits2-0.vercel.app>

■ Check Out the Source Code:

<https://github.com/rishigshshshsh-lab/VeilSplits2.0>

■ Contact details:

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The Ask

We are seeking:

- * Feedback from Stellar developers on our Soroban smart contracts.
- * Beta testing feedback from DAOs & freelancers using Stellar Testnet.
- * Ecosystem connections to Stellar Anchors & payment gateways.
- * Reviewer feedback for the next Level 6 stage upgrade.